CSC1042 Computer Network II

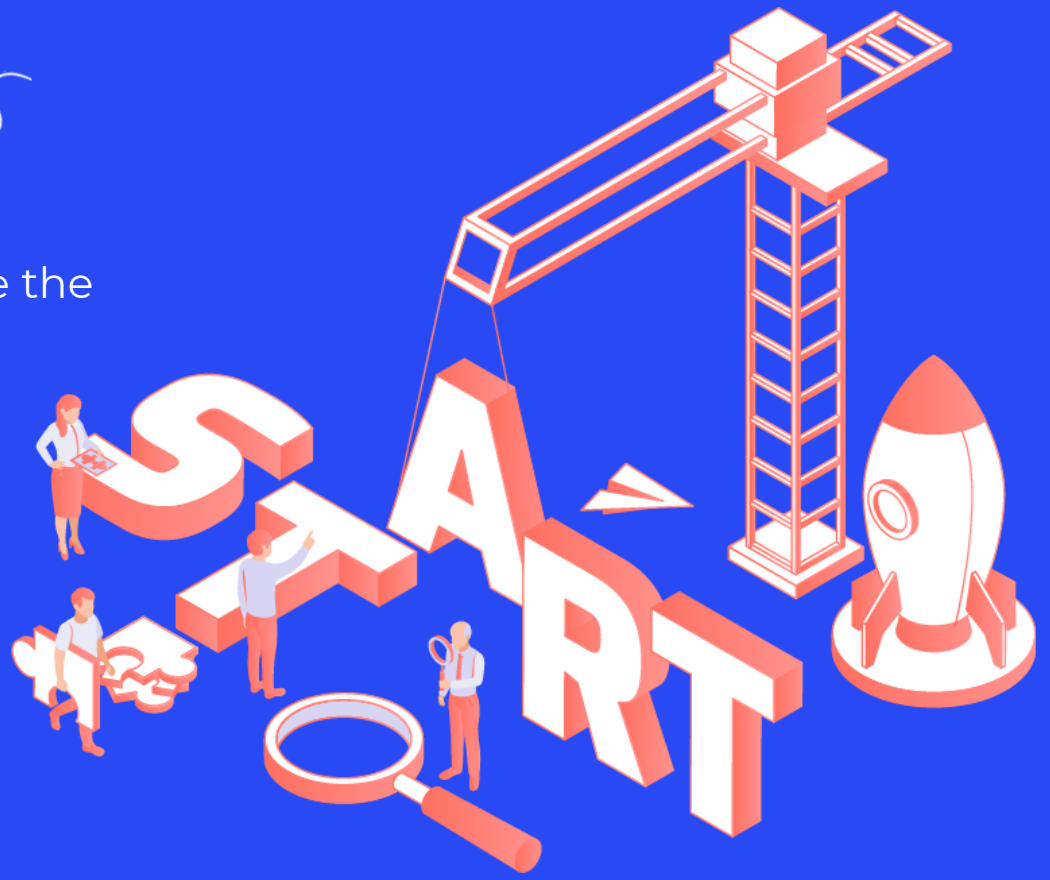
Bellman-Ford Algorithm

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Learning outcomes

- Understand how Bellman-Ford algorithm compute the shortest paths from a node to all nodes in a graph

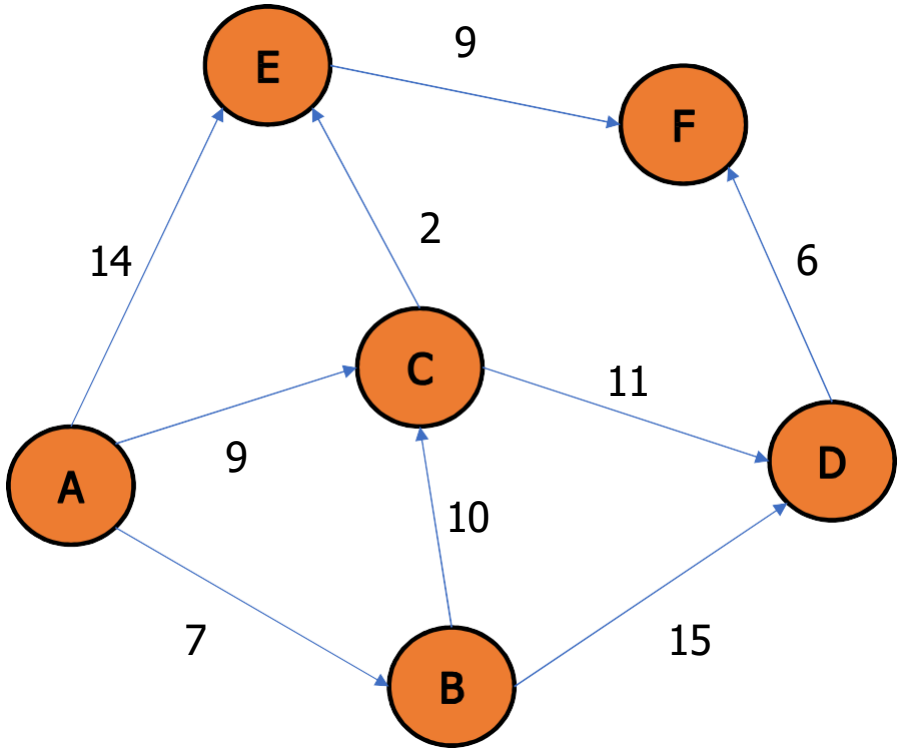


Bellman-Ford algorithm

- It allows to find the shortest paths from a node to all nodes on a graph
- Input: the graph along with start node
  - Edges on the graph are weighted
  - Can handle edges with negative weights
- Output: the shortest paths between the start node and the remaining nodes in the graph

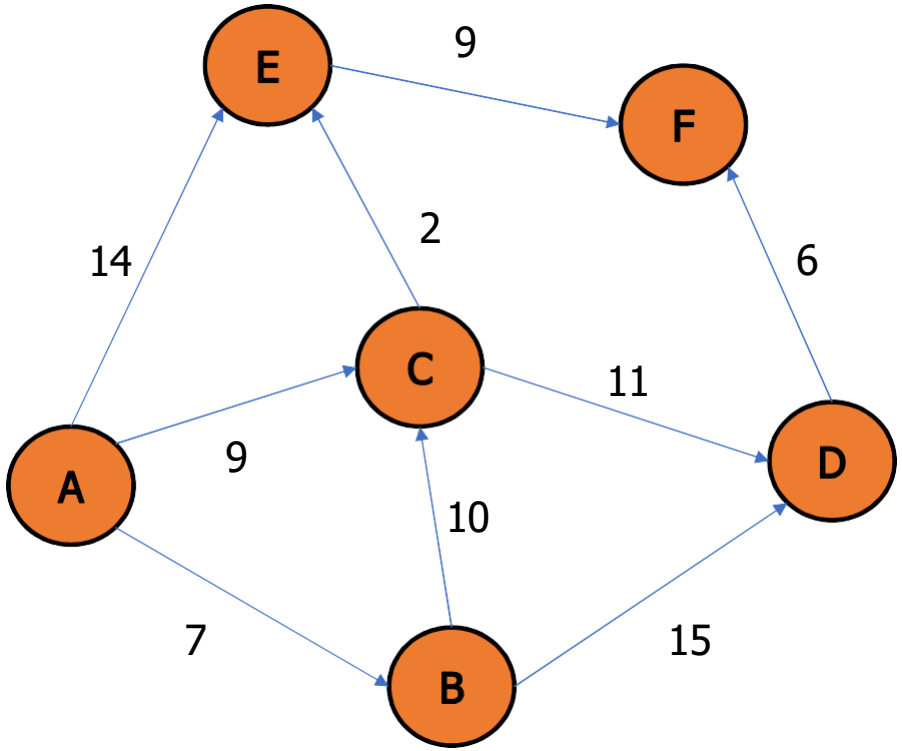
Shortest paths

- Let's look at this graph
- What is the shortest path from node A to all nodes in the graph?



Bellman-Ford algorithm

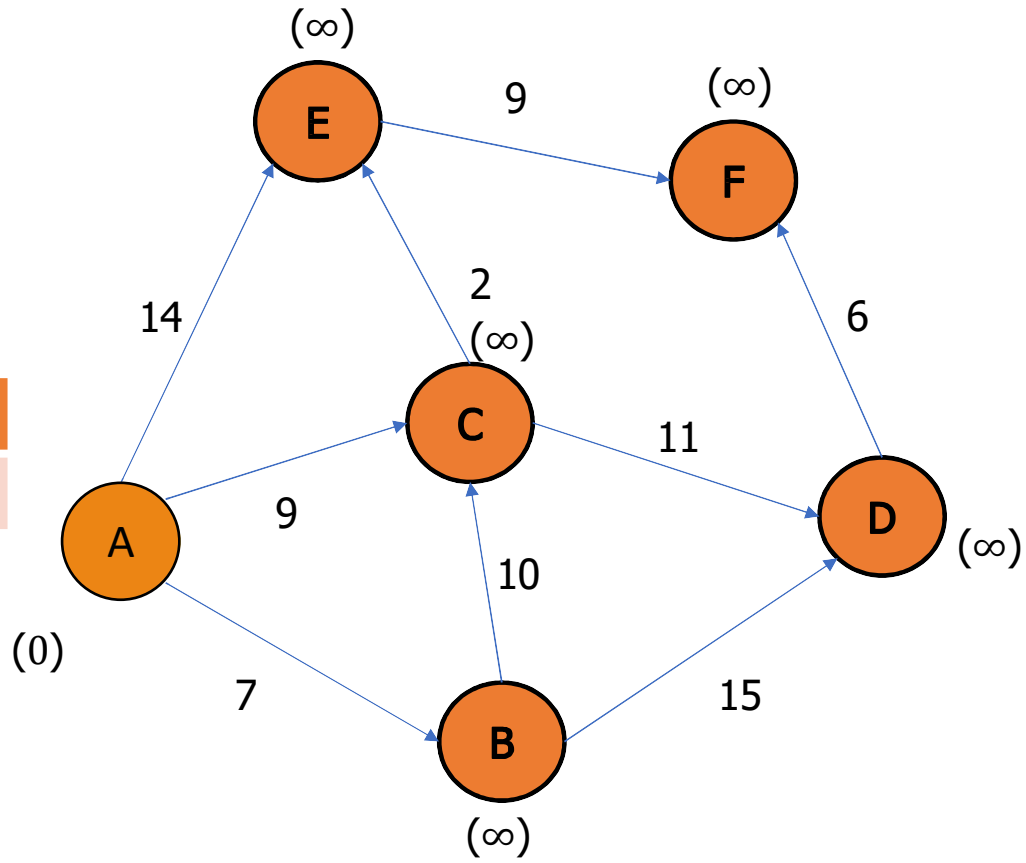
- Step 1: initialise node A with distance 0 and remaining nodes with distance ∞
- Step 2: run up to |v| - 1 iterations , where v is the number of vertices
  - For each iteration, examine all the edges in the graph
  - Look at each node, one by one, and examine its edges



Bellman-Ford algorithm

- Step 1: initialise node A with distance 0 and remaining nodes with distance ∞

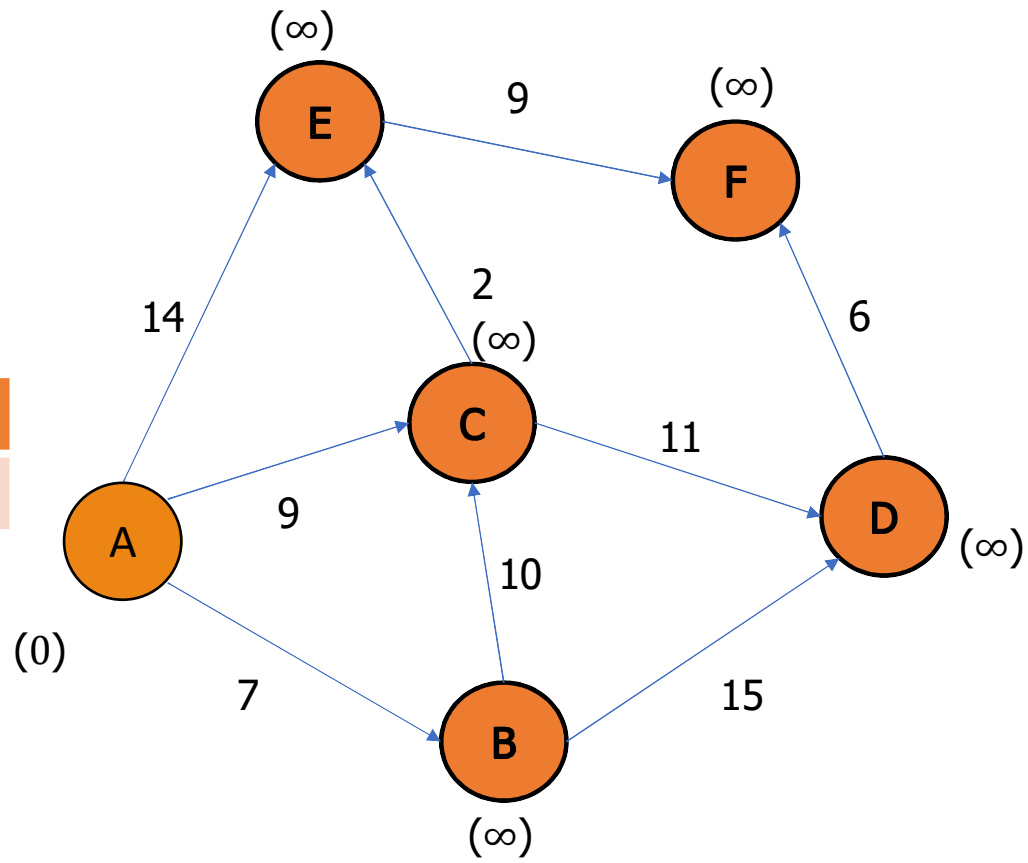
	A	B	C	D	E	F
A	0	∞	∞	∞	∞	∞



Bellman-Ford algorithm

- Step 2: since there are 6 vertices, there will be at most 6 – 1 = 5 iterations

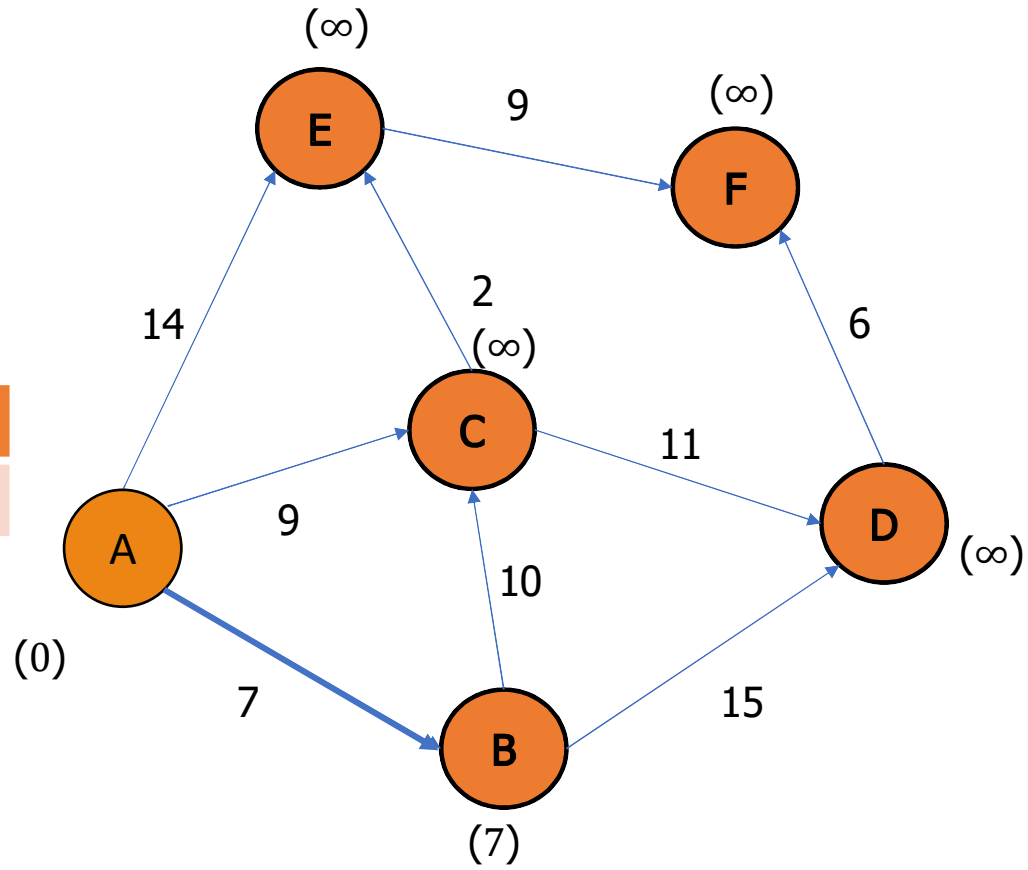
	A	B	C	D	E	F
A	0	∞	∞	∞	∞	∞



Bellman-Ford algorithm

- 1st iteration
  - Start with node A and update distances to its neighbours

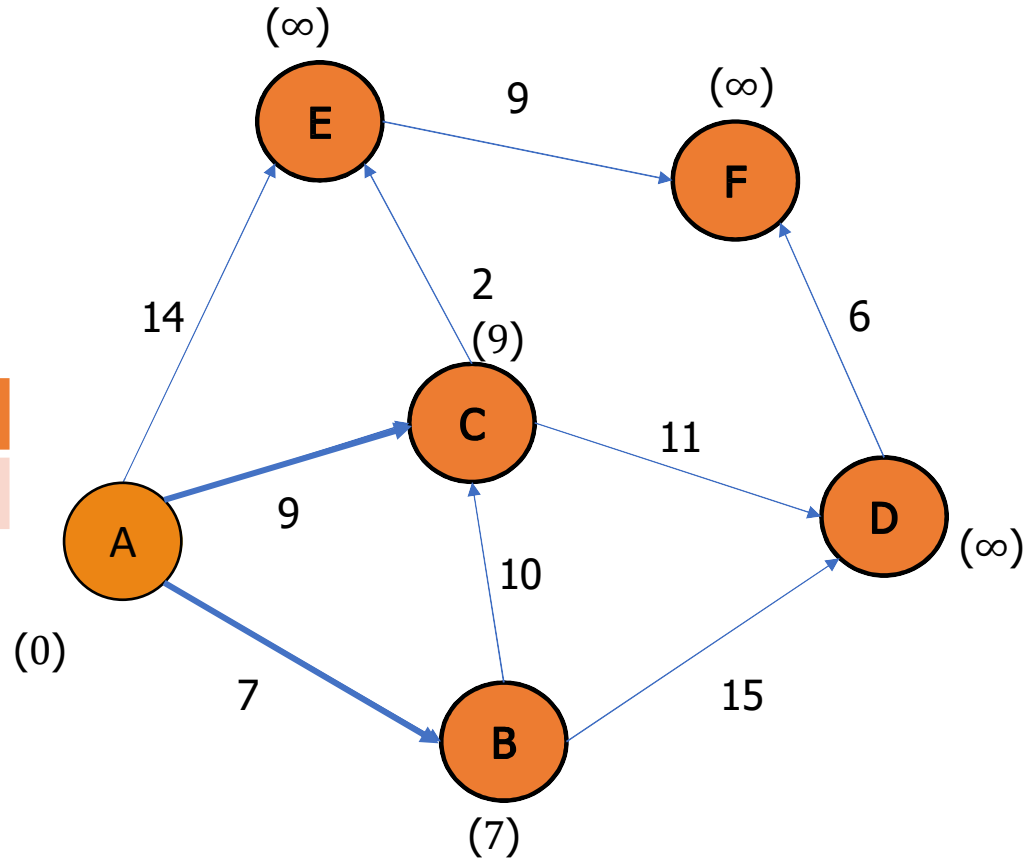
	A	B	C	D	E	F
A	0	7	∞	∞	∞	∞



Bellman-Ford algorithm

- 1st iteration
  - Start with node A and update distances to its neighbours

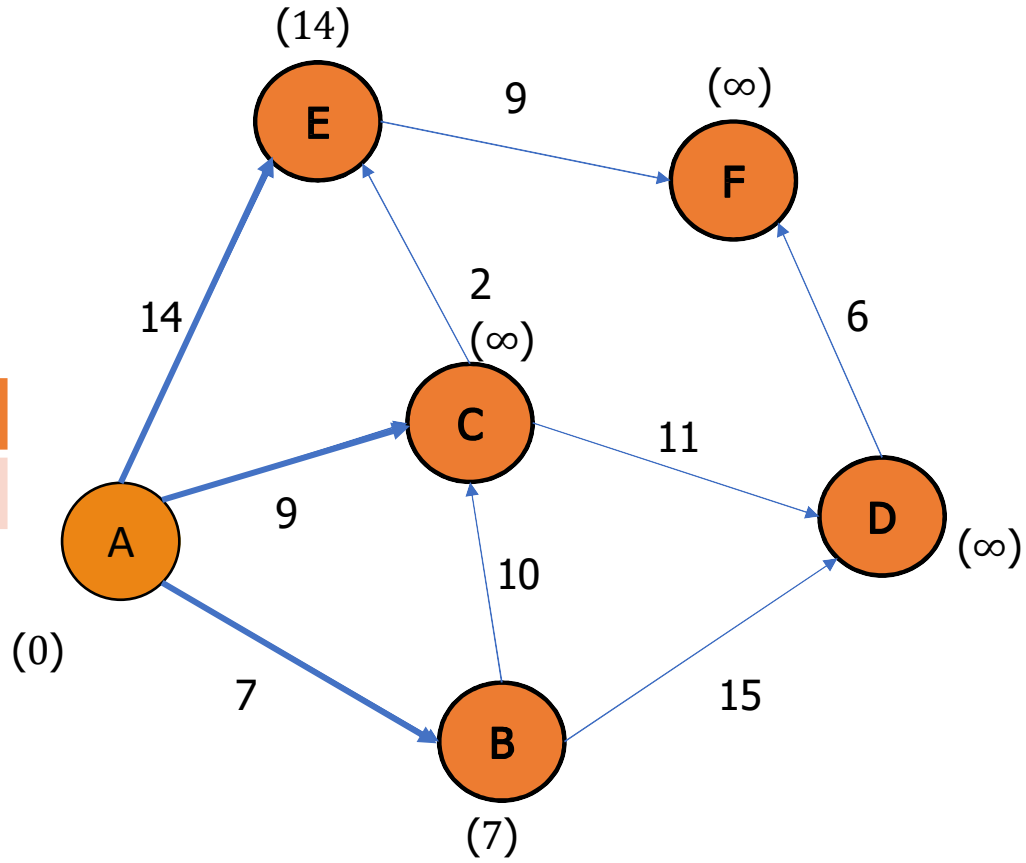
	A	B	C	D	E	F
A	0	7	9	∞	∞	∞



Bellman-Ford algorithm

- 1st iteration
  - Start with node A and update distances to its neighbours

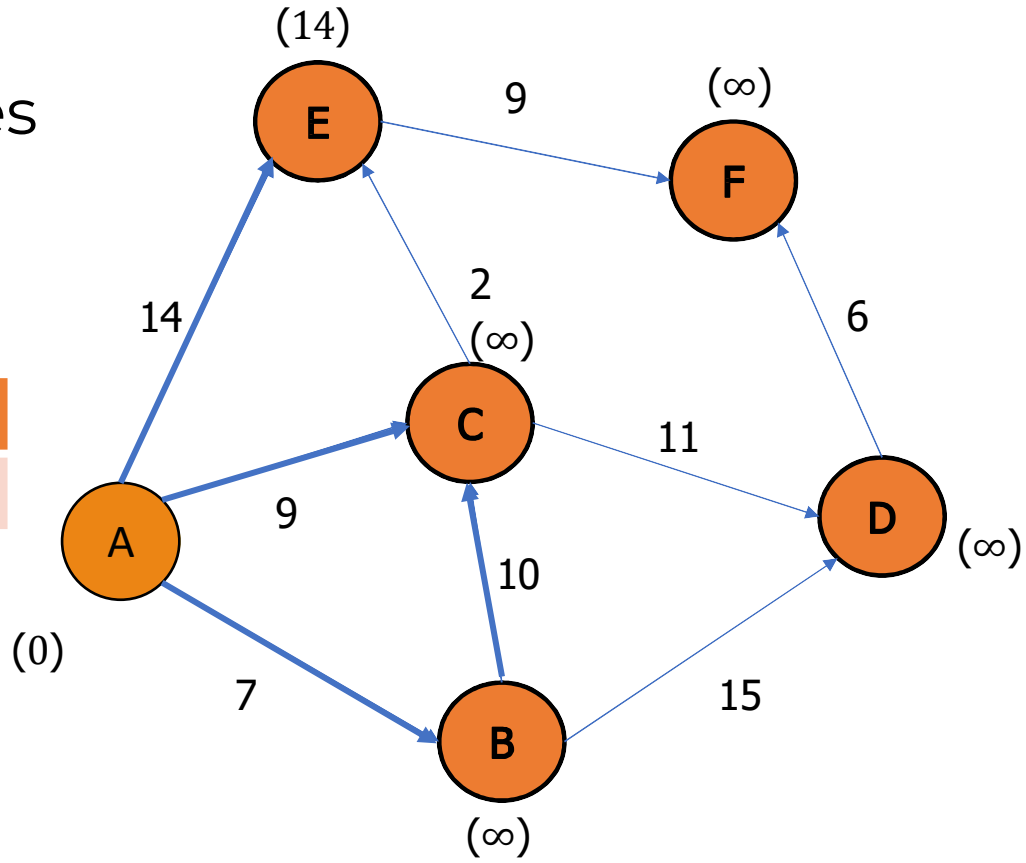
	A	B	C	D	E	F
A	0	7	9	∞	14	∞



Bellman-Ford algorithm

- 1st iteration
  - Move to node B and update distances to its neighbours

	A	B	C	D	E	F
A	0	7	9	∞	14	∞

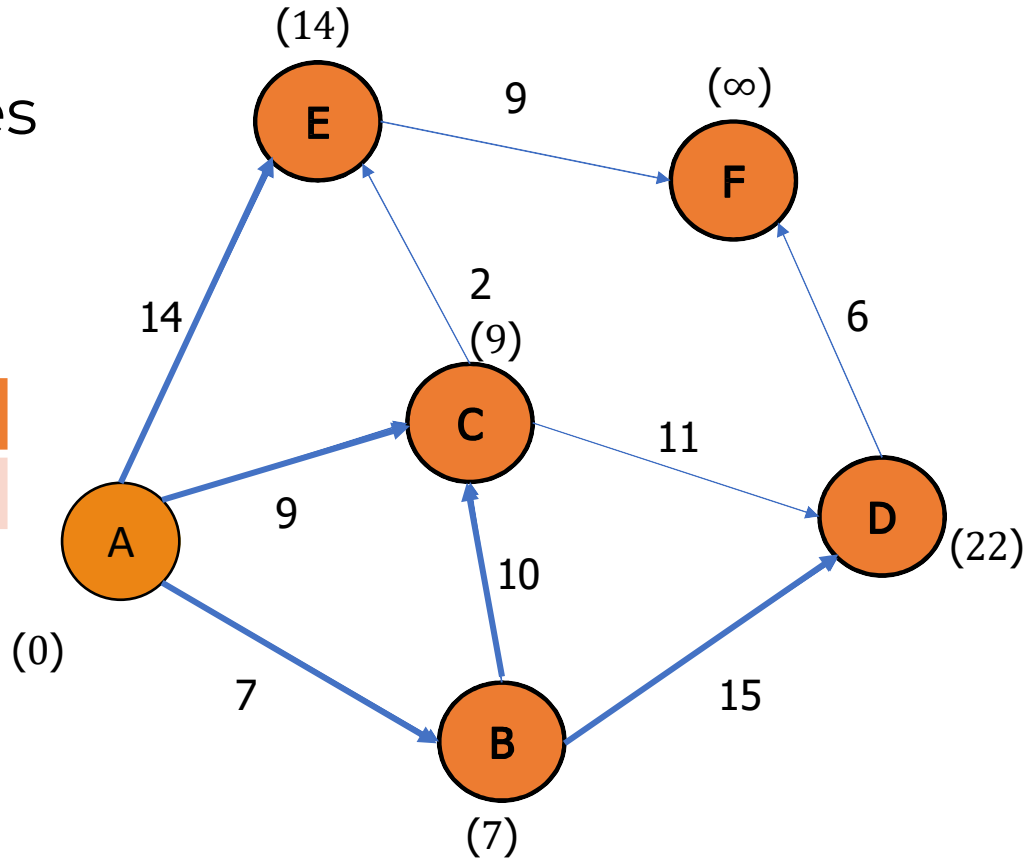


- Since the path from A to C through B has a distance of 17 > 9, we will not change C's distance

Bellman-Ford algorithm

- 1st iteration
  - Move to node B and update distances to its neighbours

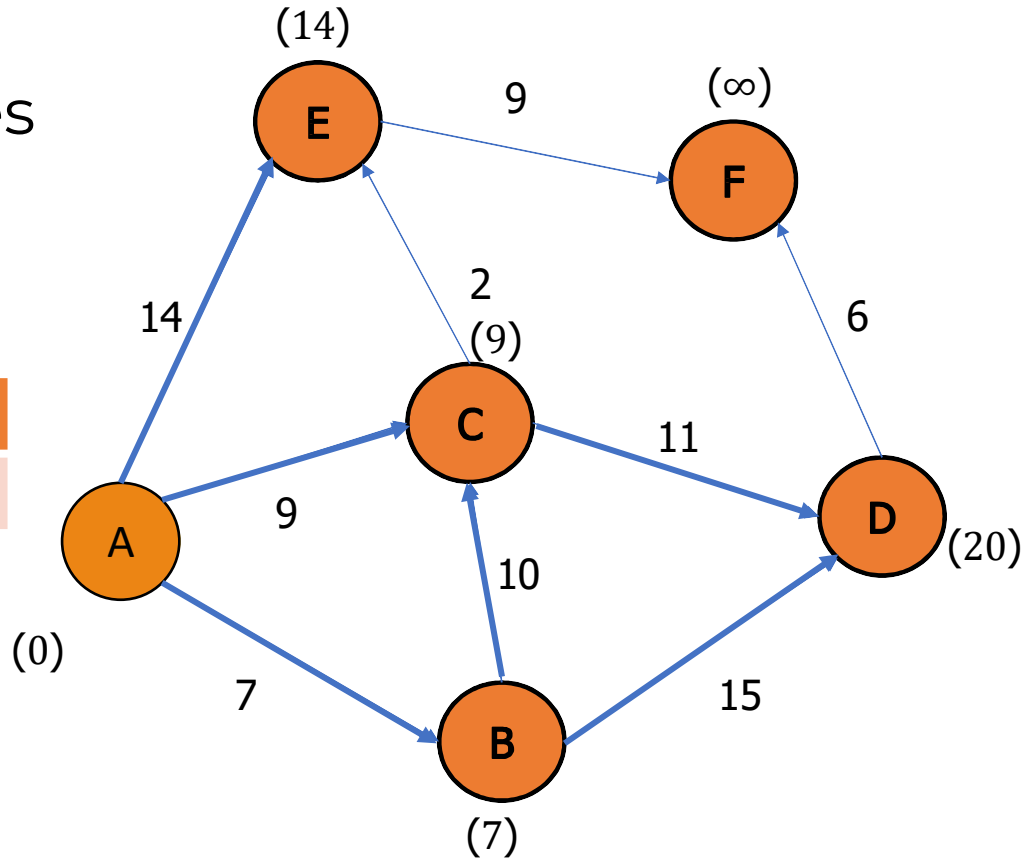
	A	B	C	D	E	F
A	0	7	9	22	14	∞



Bellman-Ford algorithm

- 1st iteration
  - Move to node C and update distances to its neighbours

	A	B	C	D	E	F
A	0	7	9	20	14	∞

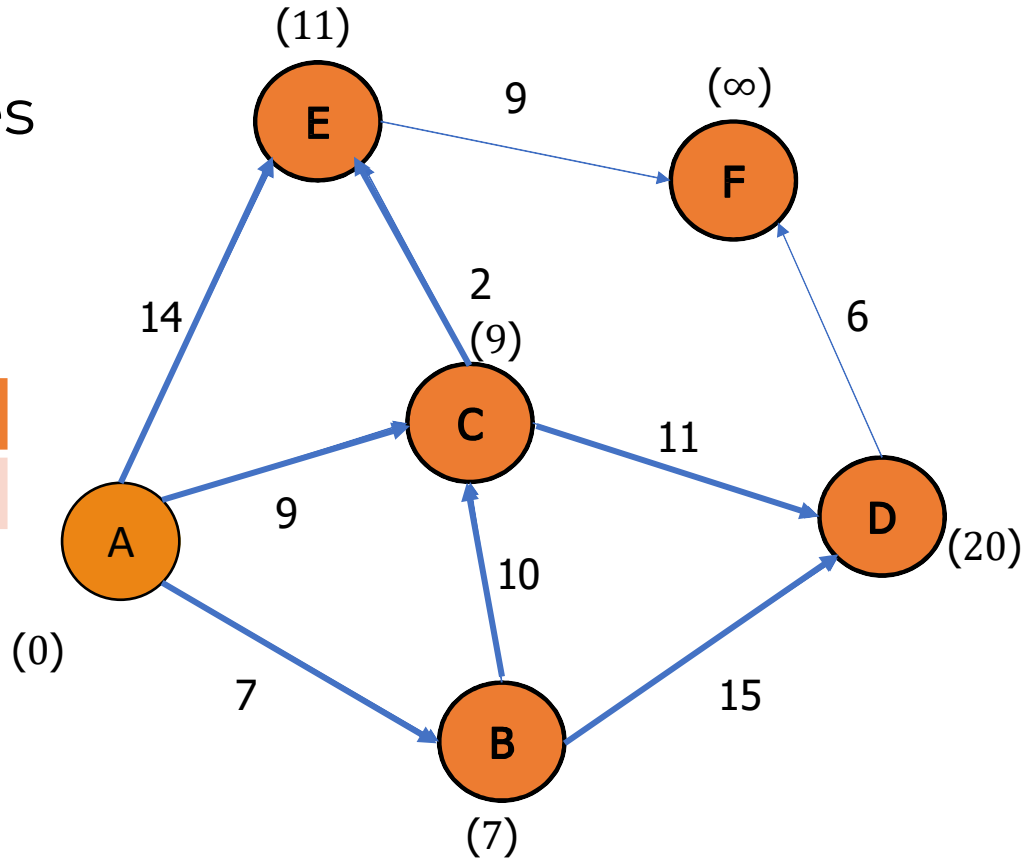


Since the path from A to D through C has a distance of 20 < 22, we update D's distance

Bellman-Ford algorithm

- 1st iteration
  - Move to node C and update distances to its neighbours

	A	B	C	D	E	F
A	0	7	9	20	11	∞

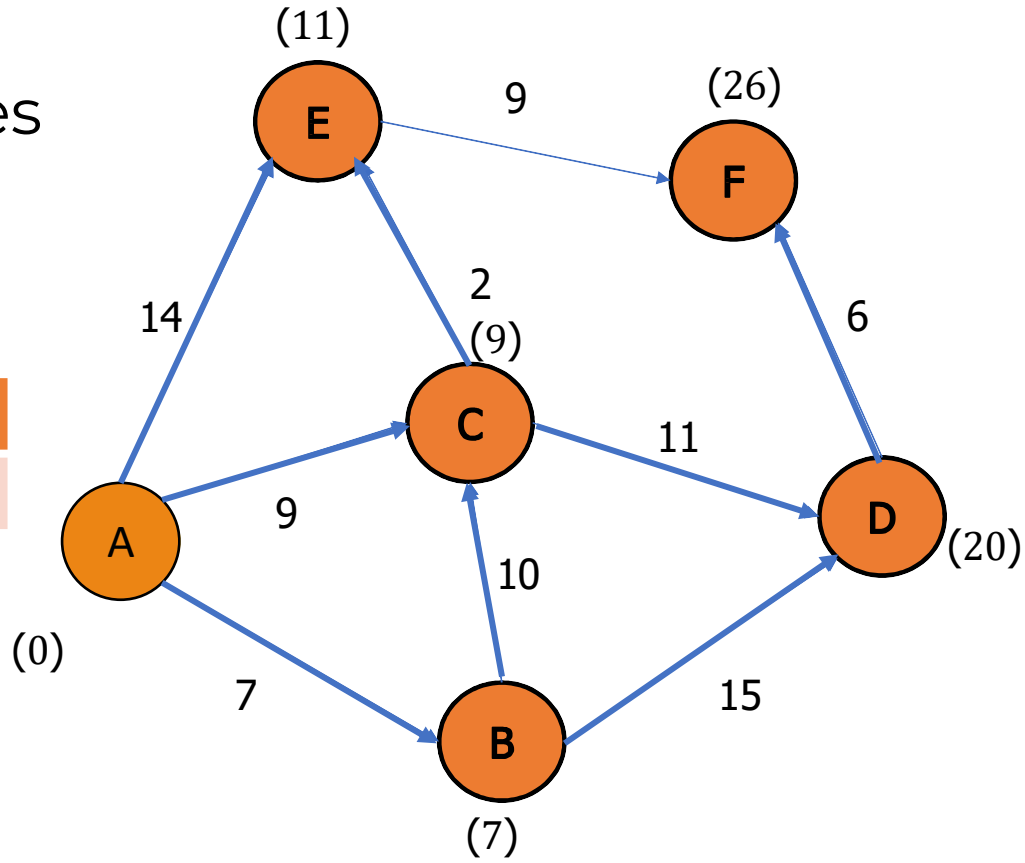


Since the path from A to E through C has a distance of 11 < 14, we update E's distance

Bellman-Ford algorithm

- 1st iteration
  - Move to node D and update distances to its neighbours

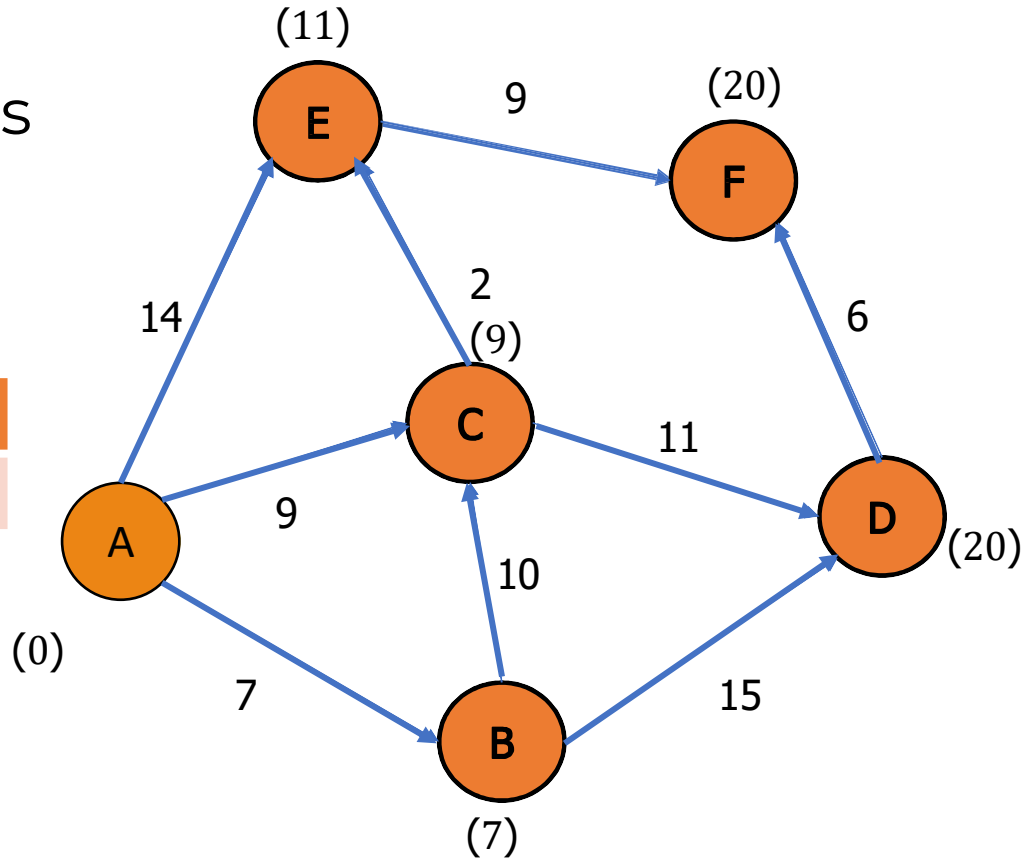
	A	B	C	D	E	F
A	0	7	9	20	11	26



Bellman-Ford algorithm

- 1st iteration
  - Move to node E and update distances to its neighbours

	A	B	C	D	E	F
A	0	7	9	20	11	20

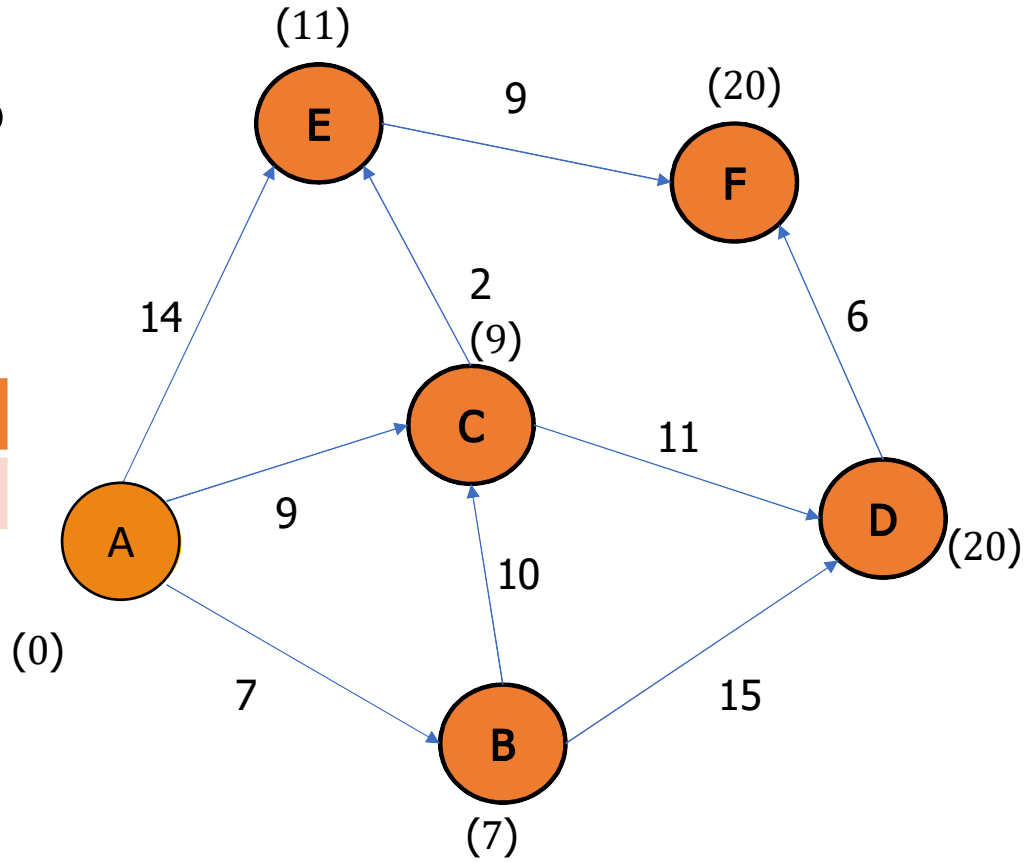


Since the path from A to F through E has a distance of 20 < 26, we update F's distance

Bellman-Ford algorithm

- 2nd iteration
  - Start from A and update distances to its neighbours and repeat the same process as in 1st iteration

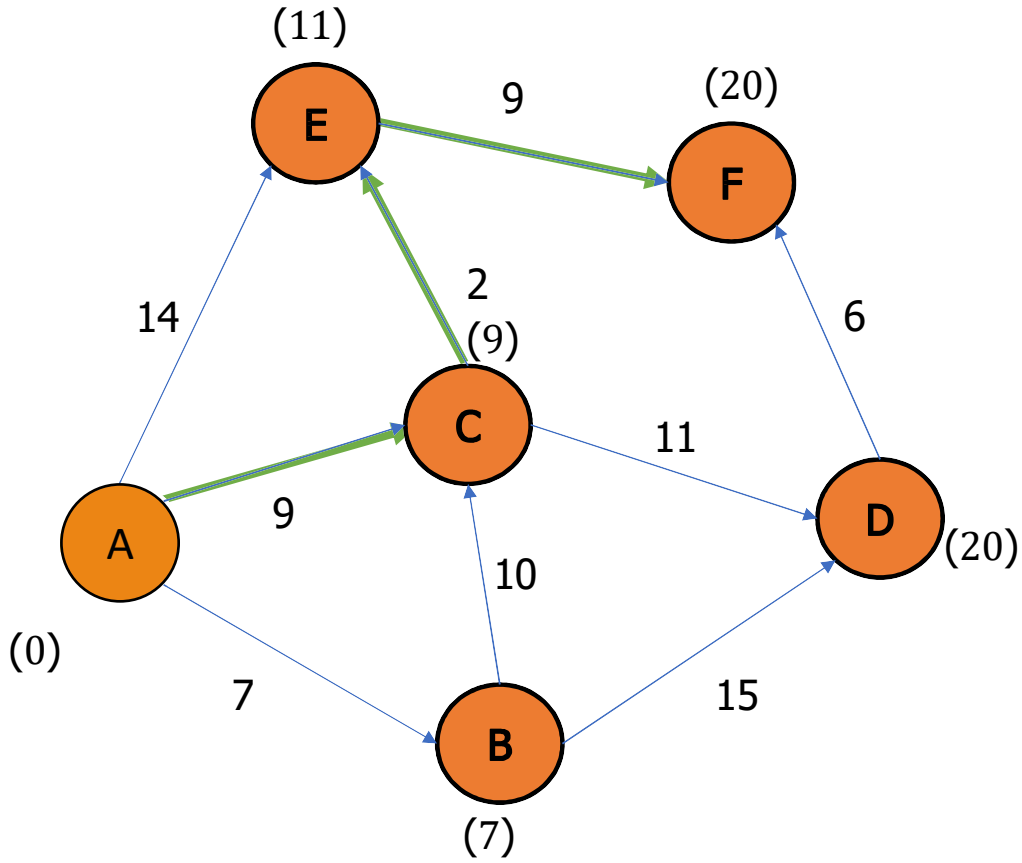
	A	B	C	D	E	F
A	0	7	9	20	11	20



- Once you checked all the edges in the graph, you will notice that no further changes can be made. Hence, the algorithm can be stopped

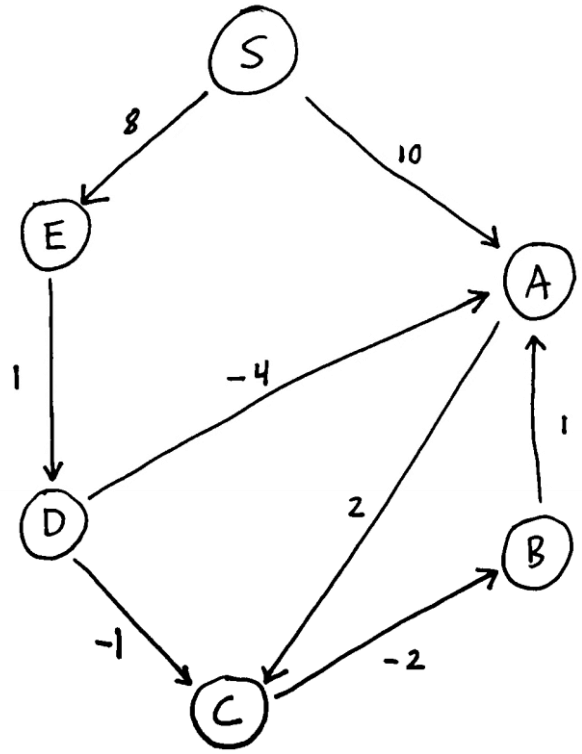
Bellman-Ford algorithm

- Therefore, the shortest path from A to F is A→C→E→F with a distance of 20



Exercise

- Apply the Bellman-Ford algorithm to find the shortest paths from S to all other nodes in the graph



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THANK YOU